Simple Voice Control

1. Function Description

Control RGB lights, buzzers, and robotic arm through the voice recognition and broadcast module.

2. Startup and Operation

2.1 Startup

Open the terminal and enter the following command to start:

```
python3 ~/dofbot_voice/scripts/simple_voice_ctrl.py
```

2.2 Operation Steps

After the program runs, the voice module will broadcast "I'm here", then say "Hi, yahboom" to the voice module. The voice module replies "I'm here" indicating successful wake-up. Then say "Turn on the red light" to the voice module. At this time, the LED light on the driver board will light up red, and the voice module will broadcast "OK, red light is on". The voice commands that this program can recognize are as follows:

Voice Command	Effect
RED-LIGHT-UP	LED light on driver board lights up red
GREEN-LIGHT-UP	LED light on driver board lights up green
BLUE-LIGHT-UP	LED light on driver board lights up blue
WARNING	Buzzer beeps 3 times
LIFT-THE-ARM-UP	Robotic arm extends upward
PUT-THE-ARM-DOWN	Robotic arm extends downward
ARM-LEFT	Robotic arm turns left
ARM-RIGHT	Robotic arm turns right
CLAMP-THE-DIP	Gripper tightens
OPEN-THE-DIP	Gripper releases

3. Core Code Analysis

`Source code path:`

```
~/dofbot_voice/scripts/simple_voice_ctrl.py
```

simple_voice_ctrl.py

```python
# Import robotic arm low-level control library
import Arm_Lib
# Import voice recognition and broadcast library
from Speech_Lib import Speech
# Create voice recognition and broadcast object mySpeech
mySpeech = Speech()
# Create robotic arm low-level control object Arm
Arm = Arm_Lib.Arm_Device()

# Call speech_read function to get voice recognition result
result = mySpeech.speech_read()

# Call Arm_RGB_set function to control LED light to turn red
Arm.Arm_RGB_set(50, 0, 0)

# Call Arm_Buzzer_On function, pass parameter 1, control buzzer to beep for 100
milliseconds
Arm.Arm_Buzzer_On(1)

# Call Arm_serial_servo_write function, pass parameters, control servo 2 to
rotate to 90 degrees within 1000 milliseconds
Arm.Arm_serial_servo_write(2, 90, 1000)

# Call void_write function, pass parameter 11, broadcast corresponding audio
file
mySpeech.void_write(11)
```